

Personal statement

Stone Chou

I am a third-year undergraduate at Stony Brook University majoring in Mathematics and Physics. My long-term goal is to pursue a research career in theoretical nuclear physics, with a focus on many-body statistics and quantum chaos. I am applying to the summer program in Physics at the University of Tokyo because of its strengths in nuclear theory that aligns closely with my current training and the direction I want to develop. In particular, I am eager to explore research with Professor Haozhao Liang on nuclear theory. Nevertheless, it is also possible for me to contribute to neutrino physics with the Nakajima's group.

Currently I am interested in working with Professor Haozhao Liang on Nuclear Theory. I hope I could get an opportunity to learn something and contribute to the University of Tokyo.

During the last two semesters, I took advanced courses to prepare for nuclear theory research. This includes two semesters of Graduate Quantum Field Theory by Professor Zamolodchikov, group theory, Quantum Hall effect, and Chern-Simons theory courses. These theoretical frameworks make it easier for me to access the topic of research.

I noticed that Professor Liang is working on the density function and the statistical properties of the heavy nucleus. Specifically, I read his work on the global prediction of nuclear charge density distribution using neural networks and found it both interesting and promising. I am currently working with Professor Verbaarschot in my university on Random Matrix Theory(RMT). We are applying GOE and GUE, which are ensembles in RMT, into the SYK model and evaluating the Quantum Chaotic Properties. This includes spectrum density, Out of Time Order Correlations(OTOCs) and the Lyapunov coefficients for the Chaotic system at large N limit. We developed and used a “dressed fermion propagator” and used it to compute operator growth in the large-N theory and infinite temperature limit. By investigating the operator growth in the SYK model we are able to learn more about the Majorana fermion system and the quantum butterfly effect. I think what I am doing now is pretty close to the paper that I read about from Professor Liang and I would be more than happy to learn new statistical techniques, say neural networks and apply to the nucleus problem. I would be excited to extend these skills to nuclear structure problems, learning the techniques needed to connect theory, computation, and uncertainty-aware prediction.

I am also interested in working with Professor Nakajima on Neutrino experiment physics. Last winter I had a conversation with Professor Nakajima and he showed me around his laboratory. Previously, I have experience working on the Deep Underground Neutrino Experiment, also known as DUNE detector. I conducted photon detector upgrade for the Phase 3 of the detector and currently still writing a paper on Atmospheric Neutrino reconstruction in Liquid Argon as the first author. In this project we are using new detector features and neutron topologies to improve the reconstruction of atmospheric neutrino and got a major improvement, you should be able to find a detailed description of my work in my CV. Anyway, I think both Professor Nakajima and Admission officer would be interested in hearing why I am eager to switch to HyperK rather than DUNE, now two competing next generation long baseline neutrino experiments: the reasons are twofold: firstly, it is my personal interest to live and work in Japan, which I stated previously, and Tokyo University as a prodigious academic research platform surely can provide me with the opportunity to work on the research I am truly interested in. Secondly, the political environment and the dilemma american project(DUNE) is now encountering make me give a second thought about working in North America: DUNE is facing a major funding problem and wouldn't be able

to complete the construction of near detector and maintaining facilities. This is not an individual problem but a reflection of the bad research environment here in US. I think in comparison HyperK with better government and maturer detecting techniques could provide me—as an international student— with better research environment. And meanwhile HyperK project also eager for young and creative people to contribute more. I am exactly the kind of future researchers HyperK is looking for.

Beyond research fit, I am motivated by the opportunity to train in the University of Tokyo's rigorous academic environment and to collaborate with researchers who are shaping the future of neutrino and nuclear physics. I have spent time in Tokyo previously and have long been inspired by Japan's scientific culture and by the University of Tokyo's role as a leading center for physics. If admitted, I hope to deepen my theoretical foundations, contribute to ongoing research in nuclear theory and/or neutrino physics, and prepare for a Ph.D. and a career in research.